Effective Probabilistic Neural Networks Model for Model-Based Reinforcement Learning USV

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***Abstract*— Gaussian process (GP) offers a robust solution for modeling the dynamics of unmanned surface vehicles (USV) in model-based reinforcement learning (MBRL). However, the rapidly increasing computational complexity with a large sample capacity of GP limits its application in complex scenarios that require substantial samples to cover the state space. In this article, a novel probabilistic MBRL approach, probabilistic neural networks model predictive control (PNMPC) is proposed to tackle this issue. With an iterative learning framework, PNMPC properly models the USV dynamics using neural networks from a probabilistic perspective to avoid the computational complexity associated with sample capacity. Employing this model to effectively propagate system uncertainties, a model predictive control (MPC) policy is developed to robustly control the USV against external disturbances. Evaluated by position-keeping and multiple targets-tracking scenarios on a real USV data-driven simulation, the proposed method consistently demonstrates its significant superiority in both model accuracy and control performance compared to not only GP model-based approaches but also the probabilistic neural networks-based MBRL baselines, across various scales of external disturbances.**

***Note to Practitioners*—Modelling the system dynamics and maintaining computational efficiency with a large sample set has been challenging for MBRL in the USV domain. We propose a novel neural network modeling method to capture the dynamic features of USV within an RL loop and develop a robust MPC policy based on its uncertainty propagation. Our method achieves computational complexity independent of the sample capacity and outperforms related baselines in model accuracy and control performance.**

***Index Terms*— Reinforcement learning (RL), neural networks, predictive models, marine vehicles.**

I. Introduction

AS ONE critical technology in marine science, unmanned surface Vehicles (USV) have evolved rapidly over the past few decades in academia and industry. These technologies have become a key solution in not only enhancing the efficiency and safety of marine transportation but also tackling the shortage of skilled professionals in the shipping industry. Despite their growing applications in various scenarios [1], [2], [3], [4], the current USV technologies highly depend on the prior knowledge of specific hardwares [5], manual selection of controller parameters [6] and large amounts of data that are usually costly to collect by human experts [7]. There is still a significant gap in achieving fully automatic control and adaptive learning in USV domian [8].

Reinforcement learning (RL) aims to iteratively learn optimal or suboptimal control strategies by interacting with the external environments without human intervention [9], [10], [11]. It offers an appealing prospect for the fully autonomous USV with the capability of adaptive learning. The traditional model-free RL approaches have been extensively applied to USV in various tasks including collision avoidance, target tracking and motion control [12], [13], [14], [15], [16], [17]. The most recent research trained a model-free RL agent in a simulation environment and successfully drove it in the real-world USV tracking task [18]. On the other hand, most of the traditional model-free RL approaches typically rely on a deterministic policy based on neural networks which inherently lacks the robustness against the ocean environment under various uncertain disturbances. They usually need substantial learning in simulated environments before they can be deployed in real-world scenarios with different unmanned systems [18], [19], [20].

This issue can be mitigated by model-based RL (MBRL) which parallelly learns a model of the environment. One of the most representative approaches, the probabilistic inference for learning control (PILCO) modeled the uncertain system by Gaussian process (GP) [21], [22] and achieved robust control with superior sampe-efficiency [23]. Based on the superior robustness of model predictive control (MPC) in complex systems like unmanned ground vehicles and space vehicles [24], [25], GP-MPC was proposed by integrating MPC into PILCO to deal with the dynamic and unpredictable disturbances in uncertain environments with evaluation on simulated cart-pole and double pendulum [26]. As the first attempt to implement the MBRL framework with GP model and MPC controller to USV, sample-efficient probabilistic model predictive control (SPMPC) successfully drove a regular size boat in a real ocean environment to learn target-reaching and position-keeping tasks without neither prior human knowledge nor additional training process in simulation [27], [28]. Drawing

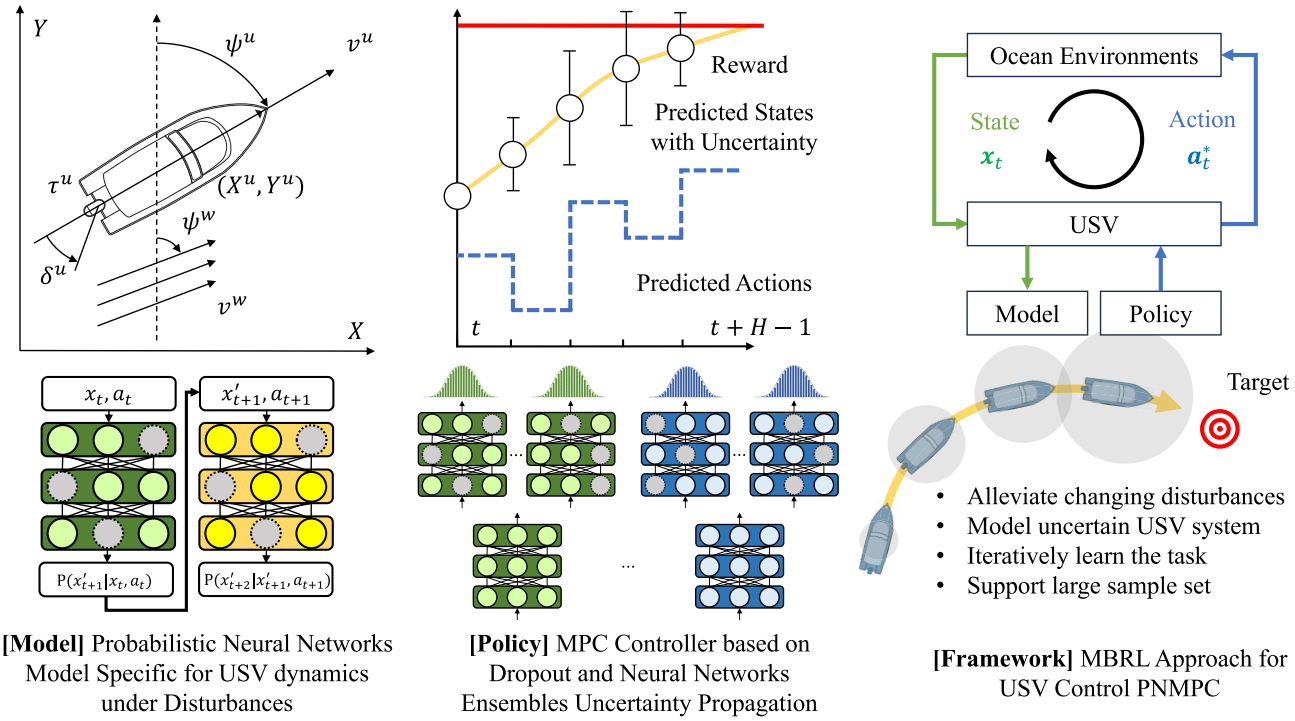
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[Model] Probabilistic Neural Networks Model Specific for USV dynamics under Disturbances

[Policy] MPC Controller based on Dropout and Neural Networks Ensembles Uncertainty Propagation

[Framework] MBRL Approach for USV Control PNMPC

Fig. 1. Principle and contributions of the proposed method PNMPC.

inspiration from [29], filtered probabilistic model predictive control (FPMPC) exploited the benefits of the probabilistic model in USV control. It employed the Bayesian filter to account for the latent uncertain variables and enhanced the control stability based on SPMPC [30].

Although SPMPC and FPMPC have shown potential in USV control, their application has been strongly limited by the non-parametric nature of GP: the computational complexity of the GP model exponentially increases at a rate of $\mathcal{O}(N^3)$ where N is the number of samples. Although this issue can be partially addressed by sparse GP [31], [32], balancing the control performance and the sample capacity of GP-based approaches remains challenging in engineering. With an independent computational complexity of the sample capacity, deep neural networks have been widely implemented to control unmanned vehicles [33], [34] and become one feasible solution to fit the probabilistic USV models. Deep Pilco first explored the power of neural networks in PILCO [35]. Probabilistic ensembles with trajectory sampling (PETS) integrated neural networks and probabilistic MPC into one MBRL framework. It achieved comparable performance to model-free RL approaches in several Mujoco control benchmarks [36]. However, despite the breakthrough in computational complexity, these approaches also exhibit inherent limitations in stability compared to their original versions based on GP. The application of them in environments with disturbances such as USV remains unexplored.

The motivation of this paper is to tackle the computational efficiency issue of traditional probabilistic MBRL approaches that employ the GP model and MPC controller in the USV domain. Designing a probabilistic neural network with sufficient stability and expressiveness, a novel MBRL method, probabilistic neural networks model predictive control (PNMPC) is proposed to properly model the uncertain USV dynamics and robustly control it against changing disturbances using deep neural networks. Our method emphasizes the features of USV dynamics in continuous time steps during the training of neural networks to improve the prediction accuracy while the unpredictable noises are further filtered in MPC for more robust and prompt control behaviors. It naturally combines the characteristics of Deep Pilco and PETS to achieve superior representative capability of system uncertainty without losing stability. Evaluated by position-keeping and targets-tracking scenarios under different levels of environmental disturbances in the real USV data-driven simulation developed by [28], PNMPC not only achieved superior control stability and prediction accuracy than the existing neural networks-based approach PETS but also comprehensively surpassed traditional MBRL methods based on GP (SPMPC, FPMPC) in control performances and model quality with an independent computational complexity to the sample capacity. The contributions of this work are summarized following Fig. 1:

1) Developing a practical probabilistic neural networks model for USV dynamics with superior generalization capability and model accuracy than both GP model and existing neural networks-based approaches.
2) Designing an effective MPC-based policy by combining the characteristics of Deep Pilco and PETS in the uncertainty propagation of probabilistic neural networks.
3) Proposing a novel MBRL approach specific for USV control by integrating the model and policy above. Its

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advantages in control capability and independent computational complexity to sample capacity compared with other baselines were evaluated in several USV scenarios.

The remainder of this paper is organized as follows. Section II provided the preliminary of the target problem. Section III detailed PNMPC. The experimental results were analyzed in Section IV. The conclusion was given in Sections V.

II. Preliminary

A. Modeling USV

In this work, the USV dynamics was described as a Markov decision process (MDP) following [28] and [30]. As shown in left top of Fig. 1, the observed state $\boldsymbol{x} \in \mathcal{S}$ was defined with elements $[X^u, Y^u, \psi^u, v^u, \delta^u, \tau^u, v^w \cdot \cos(\psi^w), v^w \cdot \sin(\psi^w)]$ where $[X^u, Y^u]$ indicate the current position of USV, ψ^u and v^u present the heading direction and velocity of USV, δ^u and τ^u are the current values of USV rudder angle and engine throttle. Both wind direction ψ^w and velocity v^w are observed by sensors while other disturbances will be treated as unobservable noises. The control action $\boldsymbol{a} \in \mathcal{A}$ included the commands of rudder angle and engine throttle $\boldsymbol{a} = [\delta^o, \tau^o]$. The superscript u, w and o represent USV, wind and operation.

This work aims to model the following USV dynamics from a probabilistic perspective where the unobservable disturbances can be well considered:

$$\boldsymbol{x}_{t+1} = \hat{f}(\boldsymbol{x}_t, \boldsymbol{a}_t). \tag{1}$$

The input state $\boldsymbol{x}_t$ and the predicted state of the next step $\boldsymbol{x}_{t+1}$ in Eq. (1) are presented in the same form for concise representation. In practice such as multiple-step prediction, we do not predict the unpredicted terms related to wind, but instead keep them consistent with the initial input state.

B. MPC-Based Policy

Based on the learned model $\hat{f}(\cdot)$, this work built an MPC controller as the policy to promptly respond to the frequently challenging disturbances following existing works [35], [36]. At time step t with the observation $\boldsymbol{x}_t$, one H-steps optimal action sequence was obtained by the MPC-based policy:

$$\begin{aligned}[\boldsymbol{a}_t^*, \ldots, \boldsymbol{a}_{t+H-1}^*] = \underset{\boldsymbol{a}_t, \ldots, \boldsymbol{a}_{t+H-1}}{\arg\max} \sum_{s=0}^{H-1} \mathbb{E}\big[\mathcal{R}(\boldsymbol{x}_{t+s+1}, \boldsymbol{a}_{t+s})\big], \\ \text{s.t. } \boldsymbol{x}_{t+s+1} = \hat{f}(\boldsymbol{x}_{t+s+1}, \boldsymbol{a}_{t+s}), \\ \boldsymbol{x}_{t+s+1} \in \mathcal{S}, \boldsymbol{a}_{t+s} \in \mathcal{A}\end{aligned} \tag{2}$$

where $\mathcal{R}(\cdot)$ is the reward function designed for a specific task. Only the first step action $\boldsymbol{a}_t^*$ was executed. The USV moved to step $t+1$ and repeated the process above. It formed a policy $\boldsymbol{a}_t^* = \pi(\boldsymbol{x}_t)$ with the capability of handling the dynamical disturbances from external environments.

An MBRL framework can be formulated by integrating the probabilistic model and MPC-based policy following [28]. The agent interacts with the ocean environment using the policy that maximizes the reward function based on its model knowledge. Its control performance is iteratively improved by updating the model through the newly obtained data.

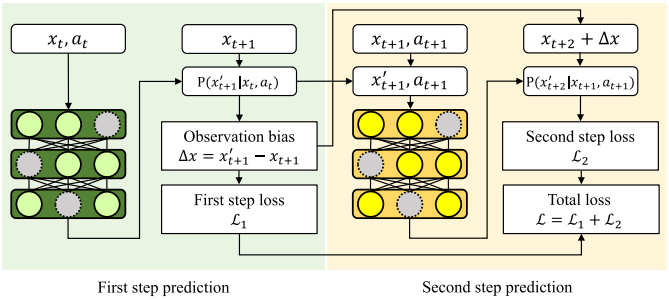


Fig. 2. Training process of the neural networks model in PNMPC.

III. Approach

A. Accurate Neural Networks Model of USV Dynamics

In this section, we first detail the process of training an effective USV model by probabilistic neural networks. The randomness of PNMPC's model mainly comes from two aspects: 1) the random dropout units of neural networks following Deep Pilco [35]; 2) multiple ensembles of neural networks with independent weights [36]. One single model of PNMPC is defined as:

$$\big[\mu(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j), \sigma(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j)\big] = \hat{f}(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j). \tag{3}$$

The networks predict both the mean and standard deviation of the state in the next step. $\boldsymbol{w}_i$ indicates one independent weights matrix from B ensembles of neural networks $\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_B\}$. $\boldsymbol{z}_j$ is one set of Q sets of dropout units $\{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_Q\}$. According to [35], $\boldsymbol{z}_j$ randomly select some dropout units in the L layers of neural networks following Bernoulli distribution $\boldsymbol{z}_j = \{\boldsymbol{z}_j^1, \ldots, \boldsymbol{z}_j^L\}$. The corresponding output at the l-th layer of the neural networks with weights $\boldsymbol{w}_i$ becomes:

$$\hat{\boldsymbol{y}}^l = \phi\big((\hat{\boldsymbol{y}}^{l-1} \circ \boldsymbol{z}_j^l)\boldsymbol{w}_i^l\big) \tag{4}$$

where $\boldsymbol{w}_i^l$ is the weights vector for the corresponding layer, $\circ$ is the element wise product, $\phi(\cdot)$ is the activation function. The one-step prediction of the probabilistic mode follows:

$$\begin{aligned}\boldsymbol{x}_{t+1} = \frac{1}{BQ} \sum_{i=1}^{B} \sum_{j=1}^{Q} \hat{f}^s(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j), \\ \hat{f}^s(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j) \sim \mathcal{N}\big(\mu(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j), \sigma(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j)\big).\end{aligned} \tag{5}$$

Although it is difficult to describe the uncertainties of the target system caused by external disturbances that follow unknown distributions, we estimate $\hat{f}^s(\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{w}_i, \boldsymbol{z}_j)$ them by Gaussian distribution according to its successful applications in ocean engineering [37].

PNMPC proposed a novel training approach of the probabilistic neural networks model with B ensembles and Q dropout sets to address the features of USV dynamics in continuous time steps. Following the main principle shown in Fig. 2, the proposed method requires the samples of USV dynamics in continuous two steps $\{\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{x}_{t+1}, \boldsymbol{a}_{t+1}, \boldsymbol{x}_{t+2}\}$.

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The loss function can be divided into two parts related to the first and second-step predictions. The first step loss follows:

$$\mathcal{L}_1 = \sum_{i=1}^{B}\sum_{j=1}^{Q}[\boldsymbol{E}_{\boldsymbol{w}_i,\boldsymbol{z}_j}^{T}\sigma^{-1}(\boldsymbol{x}_t,\boldsymbol{a}_t,\boldsymbol{w}_i,\boldsymbol{z}_j)\boldsymbol{E}_{\boldsymbol{w}_i,\boldsymbol{z}_j} + \log\det\sigma^{-1}(\boldsymbol{x}_t,\boldsymbol{a}_t,\boldsymbol{w}_i,\boldsymbol{z}_j)] \quad (6)$$

where $\boldsymbol{E}_{\boldsymbol{w}_i,\boldsymbol{z}_j} = (\mu(\boldsymbol{x}_t,\boldsymbol{a}_t,\boldsymbol{w}_i,\boldsymbol{z}_j) - \boldsymbol{x}_{t+1})$ is the prediction error of the corresponding neural networks in the first step.

The second step's loss function shares a close form of Eq. (6) with an additional observation bias $\Delta\boldsymbol{x} = \boldsymbol{x}'_{t+1} - \boldsymbol{x}_{t+1}$ where the first term is the overall prediction in Eq. (5), the second term is the next step state in samples:

$$\mathcal{L}_2 = \sum_{i=1}^{B}\sum_{j=1}^{Q}[\hat{\boldsymbol{E}}_{\boldsymbol{w}_i,\boldsymbol{z}_j}{}^{T}\sigma^{-1}(\boldsymbol{x}_{t+1},\boldsymbol{a}_{t+1},\boldsymbol{w}_i,\boldsymbol{z}_j)\hat{\boldsymbol{E}}_{\boldsymbol{w}_i,\boldsymbol{z}_j} + \log\det\sigma^{-1}(\boldsymbol{x}_{t+1},\boldsymbol{a}_{t+1},\boldsymbol{w}_i,\boldsymbol{z}_j)],$$
$$\hat{\boldsymbol{E}}_{\boldsymbol{w}_i,\boldsymbol{z}_j} = \mu(\boldsymbol{x}_{t+1},\boldsymbol{a}_{t+1},\boldsymbol{w}_i,\boldsymbol{z}_j) - \boldsymbol{x}_{t+2} - \Delta\boldsymbol{x}. \quad (7)$$

The introduced bias is added to the target state as a manual shift. It encourages the networks to focus on the prediction of the current step while reducing the negative effect of the predicted error from the previous step. The overall loss function is defined as:

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2 + \sum_{i=1}^{B}\sum_{l=1}^{L}\lambda_l\|\mathbf{w}_i^l\|_2^2 \quad (8)$$

where the third term alleviates the over-large network weights by tunable parameter vector $\boldsymbol{\lambda} = \{\lambda_1,\ldots,\lambda_L\}$. Although the prediction errors over two steps are equally considered in this work by setting the weights of $\mathcal{L}_1$ and $\mathcal{L}_2$ as 1, it is free to select different weights for specific tasks.

Compared to existing works based on GP models [28], [30] which approximate the target system by Gaussian kernel functions, PNMPC enjoys superior uncertainty expression capabilities while making fewer assumptions about noise and state distributions. Its probabilistic neural networks do not assume environmental noise follows Gaussian distributions and be able to approximate any non-linear function. The uncertainty propagation in PNMPC only requires Gaussian-distributed sampling in Eq. (5) during the model update stage, unlike moment-matching [38] which approximates input and output states as Gaussian distributions.

B. Effective Uncertain Propagation in Probabilistic MPC

In this section, we introduce an effective propagation of uncertainty in the MPC process using the probabilistic neural networks above. Given the ensembles of neural networks weights $\{\boldsymbol{w}_1,\ldots,\boldsymbol{w}_B\}$ and the dropout units set $\{z_1,\ldots,z_Q\}$, the one step prediction is divided into $B\times Q$ forward calculation following Eq. (3). During the training process introduced in Section III-A, the proposed method predicts the next step state using Gaussian distributions and incorporates the standard deviation in the loss function to guide the networks to capture the features of USV dynamics from a probabilistic perspective. In the decision-making process using the MPC-based policy, PNMPC proposed an effective uncertain propagation

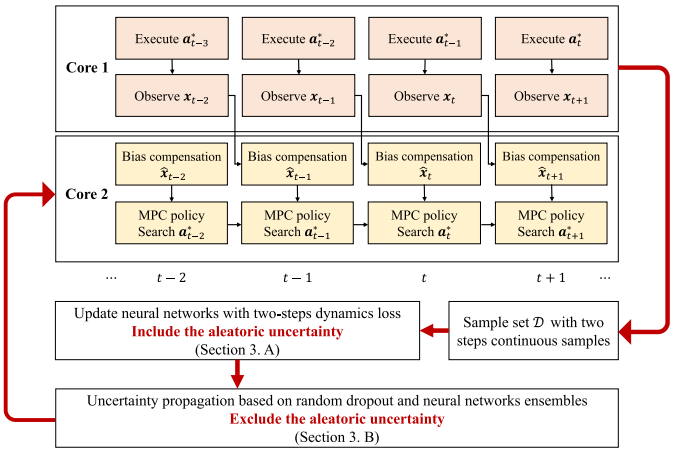


Fig. 3. Workflow of the proposed PNMPC in USV control.

that naturally combines the uncertain predictions of Deep Pilco and PETS while filtering the predicted standard deviation of each forward calculation. The MPC policy is optimized as:

$$[\boldsymbol{a}_t^*,\ldots,\boldsymbol{a}_{t+H-1}^*] = \underset{\boldsymbol{a}_t,\ldots,\boldsymbol{a}_{t+H-1}}{\arg\max}\sum_{s=0}^{H-1}\mathcal{R}(\boldsymbol{x}_{t+s+1},\boldsymbol{a}_{t+s}),$$
$$\text{s.t. } \boldsymbol{x}_{t+s+1} = \frac{1}{BQ}\sum_{i=1}^{B}\sum_{j=1}^{Q}\mu(\boldsymbol{x}_{t+s},\boldsymbol{a}_{t+s},\boldsymbol{w}_i,\boldsymbol{z}_j),$$
$$\boldsymbol{x}_{t+s+1}\in\mathcal{S},\boldsymbol{a}_{t+s}\in\mathcal{A}. \quad (9)$$

Following the related application on real-world USV [28], we assume the unpredicted wind direction and velocity do not change by fixing $\cos(\phi^w)$ and v^w in $\boldsymbol{x}$ from the first step of such a H steps optimization. The uncertainties propagated in Eq. (9) are determined by B ensembles of neural networks with independent weights and Q sets of dropout units while the predicted standard deviation of each forward calculation is blocked in the multi-step prediction. This process not only reduces the computational complexity by removing $B\times Q$ times random sampling but also mitigates the excessive amplification of standard deviation in multi-step prediction caused by the insufficiently learned model which may lead to a deteriorating control sequence in the MPC-based policy. Compared to existing works based on GP model [28], [30], it further relaxes the assumption of moment-matching [38] that predicted states must follow Gaussian distributions at the stage of decision-making. From the perspective of the source of uncertainty, the MPC-based policy above excludes the aleatoric uncertainty, i.e., the predicted standard deviation $\sigma(\cdot)$ caused by data noises and external disturbances which has a weaker Markov property related to the USV dynamics.

C. Probabilistic Neural Networks Model Predictive Control

In this subsection, we detailed the workflow of PNMPC in controlling a USV with known dynamics. The interaction was abstracted in Fig. 3. The bottom part of Fig. 3 demonstrates the training process. From right to left, the newly explored samples were stored in the sample set $\mathcal{D}$. The samples of continuous

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Algorithm 1 Learning Process of PNMPC in USV

Input sample set $\mathcal{D}$, number of network ensembles B, number of dropout units set Q, executing time Δt, horizon of MPC policy H, reward function $\mathcal{R}$.
Initialize weights ensembles and random dropout set
$\boldsymbol{w} = \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_B\}$, $z = \{z_1, \ldots, z_Q\}$
Train neural networks model by $\mathcal{D}$ with warmup samples following the loss function in Eq. (8)
$\hat{f}$ = `Train_NNS` $(\mathcal{D}, \mathcal{L}, \boldsymbol{w}, z)$
for $i = 1, 2, \ldots, N_{trial}$ **do**
  # Initialize USV and control signal
  $\boldsymbol{x}_0$ =`Reset_USV_State()`
  $\boldsymbol{a}_0^* = [0, 0]$
  for $t = 1, 2, \ldots, L_{rollout}$ **do**
    # Core 1
    `Operate_Control` $(\boldsymbol{a}_{t-1}^*, \Delta t)$
    $\boldsymbol{x}_t$ = `Observe_USV_State()`
    if $t > 1$ **then**
      # Expand $\mathcal{D}$ with two steps' samples
      $\mathcal{D} = \{\mathcal{D}, \{\boldsymbol{x}_{t-2}, \boldsymbol{a}_{t-2}^*, \boldsymbol{x}_{t-1}, \boldsymbol{a}_{t-1}^*\}\}$
    # Core 2
    # Bias compensation following Eq. (5)
    $\hat{\boldsymbol{x}}_t = \frac{1}{BQ}\sum_{i=1}^{B}\sum_{j=1}^{Q}\hat{f}^s(\boldsymbol{x}_{t-1}, \boldsymbol{a}_{t-1}, \boldsymbol{w}_i, z_j)$
    # search optimal actions following Eq. (9)
    $\boldsymbol{a}_t^*$ = `MPC_Policy` $(\hat{\boldsymbol{x}}_t, \mathcal{R}, H, \boldsymbol{w}, z)$
  # Train neural networks model by $\mathcal{D}$ following the loss function in Eq. (8)
  $\hat{f}$ = `Train_NNS` $(\mathcal{D}, \mathcal{L}, \boldsymbol{w}, z)$
Return $\hat{f}$ with $\boldsymbol{w}$, z

two steps $\{\boldsymbol{x}_t, \boldsymbol{a}_t, \boldsymbol{x}_{t+1}, \boldsymbol{a}_{t+1}\}$ were utilized to train the neural networks in a supervised learning way following Section III-A while the loss function $\mathcal{L}$ fully considered randomness from multiple ensembles of networks, different dropout sets and the aleatoric uncertainties in neural networks. The updated neural networks affected the MPC policy via Eq. (9) introduced in Section III-B, the aleatoric uncertainties of neural networks were excluded for more stable control behaviors.

The interaction process between PNMPC and the USV system mainly followed the two CPU cores setting proposed in [28], as demonstrated in the top part of Fig. 3. Two cores work in parallel to reduce the delay caused by the optimization of the MPC policy. At the beginning of each trial, the probabilistic neural networks model was initially trained by the sample set $\mathcal{D}$ with some warmup date (e.g., USV trajectories with random control signals). The RL training process can be divided into N_{trial} rollouts, each rollout has L_{rollout} steps. At the beginning of each rollout, we first reset the state of USV and obtained the initial state $\boldsymbol{x}_0$, the initial action is set as $\boldsymbol{a}_0^* = [0, 0]$. The first core was responsible for communicating with the USV system. At time step t, it sends the control signal $\boldsymbol{a}_{t-1}^*$, which is optimized by the MPC policy in the previous time step, to the USV system and receives the current state $\boldsymbol{x}_t$ from sensors. The observed state will be sent to another core at the beginning of the next step. For step $t > 1$, the samples of two continuous steps will be added to the sample set $\mathcal{D} = \{\mathcal{D}, \{\boldsymbol{x}_{t-2}, \boldsymbol{a}_{t-2}^*, \boldsymbol{x}_{t-1}, \boldsymbol{a}_{t-1}^*\}\}$.

At the same time, the second core predicts the current state $\hat{\boldsymbol{x}}_t$ based on the previous state $\boldsymbol{x}_{t-1}$ and action $\boldsymbol{a}_{t-1}$ sent by the first core following Eq. (5). Setting it as the initial state of the MPC policy effectively alleviates the bias caused by the USV operation during the optimization process, as stated in [28]. The control signal for the next step is then decided following Eq. (9) and sent to the first core at the end of this step. After each rollout with L_{rollout} steps, the model will be iteratively updated based on the current sample set $\mathcal{D}$ following the loss function $\mathcal{L}$ defined in Eq. (8). The whole learning process was summarized in Algorithm 1.

IV. Experiments

A. Experimental Settings

1) Simulation Environment: To evaluate our approach, we utilized the simulation environment developed by Furuno Electric CO. LTD., Japan which is based on a real boat (a Nissan JoyFisher with a length of seven meters) driving data in a real ocean environment. Following the dynamic model detailed in [28], the simulated boat's steering rudder had a range $\delta^o \in [-30, 30]°$, its engine throttle was mapped to $\tau^o \in [-1, 1]$, corresponding to the actions of moving backward and forward, with maximum 800 rpm in rotation speed. The executing time was set as $\Delta t = 2.0$ second.

There are two types of disturbances in this simulation: one is the wind with speed v^w and direction ψ^w that can be detected by sensors, and the other is the unobservable wave with speed v^c and direction ψ^c. Following the settings in related works [28], [30], three levels of rand wind and current velocity were set to simulate the ocean environments under disturbances:

1) $v^w = 2.0 + U(-0.1, 0.1)$, $v_c = 0.25 + U(-0.1, 0.1)$ m/s,
2) $v^w = 4.0 + U(-0.1, 0.1)$, $v_c = 0.5 + U(-0.1, 0.1)$ m/s,
3) $v^w = 6.0 + U(-0.1, 0.1)$, $v_c = 0.5 + U(-0.1, 0.1)$ m/s.

U indicates a uniform distribution. The increasing wind and current speeds led to a significantly enhanced uncertainty of USV dynamics and made the learning of robust policy more challenging. Both wind and current directions were initialed by a certain value and affected by random noises at each step:

$$\psi^w = 37° + U(-30, 30)°, \psi^c = 100° + U(-30, 30)°.$$

2) Control Scenarios: Two control scenarios: position-keeping and multiple targets-tracking were conducted. The first task focused on staying the USV near its initial position. Its reward function was defined as:

$$\mathcal{R}(\boldsymbol{x}_{t+1}) = -||\boldsymbol{p}_{t+1} - \boldsymbol{p}_{\text{target}}||^2. \tag{10}$$

$\boldsymbol{p}_{t+1}$ indicated the USV position $[X^u, Y^u]$ at step $t+1$, $\boldsymbol{p}_{\text{target}} = [0, 0]$ was the initial position. The learning process in Algorithm 1 was conducted with $N_{\text{trial}} = 20$ iterations/rollouts, each rollout contains $L_{\text{rollout}} = 50$ steps. After training, the agent will be tested by 30 rollouts with $L_{\text{rollout}} = 100$ steps. In addition to the offset from the target position, we defined two additional evaluation criteria. The final success rate is determined by the ratio of staying USV near the target within a 7 m range at the end of the test, The overall success rate is

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TABLE I
COMPARISON OF PROPOSED METHOD AND RELATED BASELINES

Approach	Model type	Uncertainty propagation	Horizon of model fitting	Policy type	Computational complexity related to sample size N
DroQ [39]	N/A	N/A	N/A	Actor-Critic Networks	Independent
TD7 [40]	N/A	N/A	N/A	Actor-Critic Networks	Independent
SPMPC [28]	Gaussian process	Moment-Matching	Single step	MPC controller	$\mathcal{O}(N^3)$
FPMPC [30]	Gaussian process	Moment-Matching	Single step	MPC controller	$\mathcal{O}(N^3)$
PETS [36]	Neural networks	Ensembles sampling	Single step	MPC controller	Independent
PNMPC (ours)	Neural networks	Ensembles sampling + Random dropout	Continuous two steps	MPC controller	Independent

determined by whether the USV holds its position within the range above during the entire test rollout.

The second task focused on sequentially passing multiple targets and staying near the last one. Two trajectories were designed for this task. The "circle" one included six targets at $[15.55, 12.89]$, $[22.42, 32.15]$, $[9.83, 46.77]$, $[-13.45, 45.36]$, $[-22.31, 28.53]$, $[-13.25, 10.35]$ as one circle (120 m total distance) to test USV's control skill in local scenes. The "right" one included ten linearly spacing targets from $[0, 0]$ to $[40, 200]$ (200 m total distance) to evaluate the control behavior in a long-term distance. The reward function was defined as:

$$\mathcal{R}(\boldsymbol{p}_{t+1}) = -||\boldsymbol{p}_{t+1} - \boldsymbol{p}^{i}_{\text{target}}||^2 - 0.5 \cdot ||\boldsymbol{p}_{t+1} - \boldsymbol{p}^{i+1}_{\text{target}}||^2 \quad (11)$$

where the agent minimized its distances to the current and the next targets (denoted by index i and $i+1$) when there is more than one target. It turned to Eq. 9 with the last target. The tracking distance was defined as:

$$D(t) = \sum_{j=1}^{i} \left(||\boldsymbol{p}^{j}_{\text{target}} - \boldsymbol{p}^{j-1}_{\text{target}}||^2 \right) - ||\boldsymbol{p}_t - \boldsymbol{p}^{i}_{\text{target}}||^2. \quad (12)$$

The complete ratio d was calculated by dividing it at the last step by the total distance. The rollout was classified as a success with $d \geq 95\%$, and as a failure with $d < 60\%$. The learning process was conducted with $N_{\text{trial}} = 20$ iterations/rollouts, each rollout contains $L_{\text{rollout}} = 100$ steps. After training, the agent will be tested by 30 rollouts with $L_{\text{rollout}} = 100$ steps.

3) Baselines and Parameters: Five baselines were selected in the experiment. The SPMPC proposed by [28] is the original MBRL approach based on GP. FPMPC [30] can be treated as its SOTA extension with support of partially observable MDP. For the neural networks-based approach, PETS [36] was compared. It can be seen as the proposed method without the features of model training and uncertainty propagation introduced in Sections. III-A and III-B. Two state-of-the-art model-free RL approaches DroQ [39] and TD7 [40] were selected for a fair comparison. All compared methods were implemented by PyTorch [41]. All MPC policies were implemented based on the bound optimization by quadratic approximation [42] in NLopt[1] for a fair comparison. The settings of SPMPC and FPMPC followed their original paper, where the sparse GP [31] was used to enable an $H = 3$ steps prediction horizon in MPC. According to the discussion of computational complexity in [28], it is difficult to complete MPC's optimization within 2 seconds with $H > 3$ during a 20 iterations' learning procedure. PETS and PNMPC benefited from the independent computational complexity of the sample capability. We set $H = 10$ for them to fully demonstrate these advantages. The neural networks were built with two hidden layers with 200 units, other related parameters were as follows: $B = 5$, $Q = 4$, $\boldsymbol{\lambda} = [0.0001, 0.00025, 0.00025, 0.0005]$ for input, output and hidden layers in Eq. (8), the supervised learning of neural works were conducted in 40 epoches with learning rate 0.001. To evaluate the impact of the uncertainties in our method, we tested PNMPC without uncertainty propagation (indicated as PNMPC-N), i.e., the size of ensembles was set to $B = 1$ while the dropout units were also disabled.

Table I summarized the characteristics of PNMPC and other baselines. Leading model-free RL approaches, DroQ and TD7, learned control policies by updating Actor-Critic networks without constructing explicit models. Their computational complexity depended on the network structure rather than the sample size. In comparison, SPMPC and FPMPC modeled the target dynamics using GP and derived control policies from MPC controllers based on GP's uncertainty propagation using analytic moment matching [38], offering robust resistance to environmental disturbances. However, GP's computational complexity increased exponentially with sample size, hampering its applications on USV with large sample sets. PETS addressed this issue by substituting GP with probabilistic neural networks and estimated propagated uncertainties through network ensembles. Our method PNMPC further enhanced PETS to meet the requirement of USV: it considered the system dynamics in two continuous steps during the model learning process and integrated random dropout from Deep Pilco [35] for superior representation capability of uncertainties.

All experimental results were conducted on a computational server with an Intel Xeon(R) W-2245 CPU, NVIDIA GeForce RTX 2080 Ti and 64 GB memory by five independent trials with different random seeds.

B. Evaluation of Learning and Control Performances

1) Learning Performances: The learning curves of all compared methods within 20 iterations were compared in Fig. 4 where the Y axis indicated the average position offset in the position-keeping task and the average tracking distance

[1] http://github.com/stevengj/nlopt

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TABLE II
TEST RESULT OF PROPOSED METHOD AND BASELINES IN THE POSITION-KEEPING TASK

Disturbance	Method	Average [m]	Median [m]	Time [s]	Success rate (final)	Success rate (overall)
Lv. 1	DroQ	47.16 ± 28.55	43.96	0.002 ± 0.0003	0.0%	0.0%
	TD7	20.70 ± 15.88	29.02	$\mathbf{0.002 \pm 0.0002}$	10.67%	9.33%
	SPMPC	3.94 ± 6.71	1.51	0.43 ± 0.27	78.67%	66.0%
	FPMPC	1.85 ± 3.15	1.30	0.69 ± 0.83	**97.33%**	48.0%
	PETS	7.69 ± 10.04	4.0	0.24 ± 0.09	90.0%	3.33%
	PNMPC	$\mathbf{1.85 \pm 2.72}$	**1.14**	0.25 ± 0.08	95.33%	**80.0%**
	PNMPC-N	18.48 ± 18.03	6.41	0.19 ± 0.01	68.0%	0.0%
Lv. 2	DroQ	82.32 ± 51.42	79.23	0.002 ± 0.0003	0.0%	0.0%
	TD7	23.85 ± 27.98	16.90	$\mathbf{0.002 \pm 0.0002}$	34.0%	20.0%
	SPMPC	4.90 ± 9.98	2.93	0.36 ± 0.24	85.33%	25.33%
	FPMPC	4.32 ± 7.17	2.67	0.47 ± 0.49	90.0%	26.0%
	PETS	26.17 ± 28.24	15.65	0.25 ± 0.10	49.33%	0.67%
	PNMPC	$\mathbf{3.50 \pm 4.97}$	**1.85**	0.25 ± 0.09	**92.67%**	**51.33%**
	PNMPC-N	22.42 ± 35.07	11.41	0.19 ± 0.01	32.0%	0.0%
Lv. 3	DroQ	79.26 ± 48.21	77.18	$\mathbf{0.002 \pm 0.0003}$	0.0%	0.0%
	TD7	41.91 ± 47.49	48.98	$\mathbf{0.002 \pm 0.0003}$	27.33%	15.33%
	SPMPC	14.0 ± 28.64	3.30	0.30 ± 0.18	72.0%	20.0%
	FPMPC	5.33 ± 9.85	3.47	0.56 ± 0.50	84.67%	0.0%
	PETS	24.70 ± 20.52	18.61	0.24 ± 0.09	26.67%	0.0%
	PNMPC	$\mathbf{3.42 \pm 4.03}$	**2.04**	0.25 ± 0.09	**98.0%**	**35.33%**
	PNMPC-N	35.21 ± 30.24	26.84	0.19 ± 0.01	29.33%	0.0%

in the targets-tracking task with circle and right trajectories. The original SPMPC gradually failed to converge to proper policies as the level of disturbance increased. The neural networks-based PETS had large standard deviations in the average position offset during the training, making it difficult to converge under all levels of disturbances. As a comparison, PNMPC stably converged to the same or even superior control policies as FPMPC under all levels of disturbances and enjoyed a leading convergence velocity. In the targets-tracking scenario with both circle and right trajectories, we observed similar advantages in the learning capability with PNMPC. It usually converged to higher average tracking distances than FPMPC while the performances of SPMPC were significantly influenced by the degree of disturbances. The neural networks-based PETS struggled to learn comparable control policies in most cases without the proposed tricks in modeling USV dynamics and propagating system uncertainties. Once the uncertainty propagation was disabled, PNMPC-N exhibited a decline in both learning ability and stability, falling significantly behind other probabilistic MBRL approaches. In terms of model-free methods, DroQ and TD7 were unable to successfully learn the tasks within 20 iterations due to the perturbed environment and the limited number of interactions.

The results above demonstrated the feasibility of the proposed method in MBRL USV. Compared to the traditional neural networks-based approach PETS, our method had robustly learned superior control strategies in the MBRL loop. Its performances could have certain advantages over the mature MBRL approach based on the GP models.

2) Testing Performances: The control performances of PNMPC and other baselines in the testing were compared in Tables II, III and IV to investigate the generalization capability of our method after the training procedure. The average and median terms in Table II were related to the position offset, the time term in all tables indicated the average optimization time of the MPC policy per step. The bold font indicated the best performance in the corresponding term.

In the position-keeping task, PNMPC had shown significant advantages under different levels of disturbances over other baselines in terms of average position offset, median position offset, and task success rates in all tests, apart from the 2% less success rate (final) in the first level disturbance compared to FPMPC. As a comparison, the neural networks-based PETS performed consistently worse than SPMPC and FPMPC in position offset and success rates. For computational efficiency, the proposed method had a lower average optimization time than other GP model-based approaches while enjoying a longer prediction horizon. In the targets-tracking task with the circle trajectory, the superiority of PNMPC was observed. The proposed method outperformed other baselines in all control-related metrics. Compared to the suboptimal approach FPMPC, PNMPC improved the average tracking distance by 8% under the highest disturbance level, resulting in a 30% increase in the success rate. In contrast, the control performance of PETS and SPMPC quickly deteriorated with increasing disturbance levels. In terms of computational complexity, the GP-based method SPMPC and FPMPC had achieved certain advantages which were potentially caused by

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TABLE III
TEST RESULT OF PROPOSED METHOD AND BASELINES IN THE TARGETS-TRACKING TASK (CIRCLE)

Disturbance	Method	Ave. complete distance [m]	Ave. complete ratio	Time [s]	Success rate	Failure rate
Lv. 1	DroQ	-170.31 ± 59.80	-141.92%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-150.70 ± 94.01	-125.58%	$\mathbf{0.002 \pm 0.0002}$	0.0%	100.0%
	SPMPC	102.0 ± 18.49	85.0%	0.14 ± 0.09	50.0%	8.0%
	FPMPC	102.80 ± 20.01	85.57%	0.13 ± 0.08	56.0%	9.33%
	PETS	100.33 ± 26.95	83.61%	0.21 ± 0.02	53.33%	19.33%
	PNMPC	$\mathbf{114.61 \pm 13.61}$	$\mathbf{95.51}\%$	0.22 ± 0.10	$\mathbf{88.0}\%$	$\mathbf{4.67}\%$
	PNMPC-N	79.66 ± 47.55	66.39%	0.14 ± 0.004	40.67%	37.33%
Lv. 2	DroQ	-172.23 ± 67.71	-143.52%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-49.18 ± 41.45	-40.98%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	SPMPC	94.66 ± 28.03	79.14%	0.13 ± 0.08	56.67%	20.0%
	FPMPC	93.91 ± 21.02	78.26%	0.12 ± 0.07	31.33%	19.33%
	PETS	45.85 ± 51.78	38.21%	0.21 ± 0.10	10.0%	70.67%
	PNMPC	$\mathbf{104.81 \pm 26.14}$	$\mathbf{87.35}\%$	0.22 ± 0.11	$\mathbf{69.33}\%$	$\mathbf{14.0}\%$
	PNMPC-N	52.38 ± 39.04	43.65%	0.14 ± 0.002	10.0%	74.67%
Lv. 3	DroQ	-130.10 ± 37.68	-108.42%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-194.02 ± 221.73	-161.68%	$\mathbf{0.002 \pm 0.0002}$	0.0%	100.0%
	SPMPC	45.23 ± 82.01	37.69%	0.12 ± 0.07	13.33%	40.0%
	FPMPC	82.03 ± 22.23	68.40%	0.11 ± 0.06	18.67%	40.0%
	PETS	58.86 ± 48.87	49.05%	0.21 ± 0.10	16.67%	63.33%
	PNMPC	$\mathbf{91.49 \pm 30.89}$	$\mathbf{76.24}\%$	0.22 ± 0.10	$\mathbf{48.0}\%$	$\mathbf{29.33}\%$
	PNMPC-N	5.08 ± 82.87	4.23%	0.19 ± 0.001	5.33%	82.67%

TABLE IV
TEST RESULT OF PROPOSED METHOD AND BASELINES IN THE TARGETS-TRACKING TASK (RIGHT)

Disturbance	Method	Ave. complete distance [m]	Ave. complete ratio	Time [s]	Success rate	Failure rate
Lv. 1	DroQ	-116.35 ± 90.91	-58.17%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-51.65 ± 42.30	-25.82%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	SPMPC	182.62 ± 30.86	91.31%	0.18 ± 0.12	77.33%	14.0%
	FPMPC	190.50 ± 17.80	95.25%	0.16 ± 0.11	86.0%	$\mathbf{1.33}\%$
	PETS	133.89 ± 26.95	58.50%	0.23 ± 0.10	21.33%	40.67%
	PNMPC	$\mathbf{191.07 \pm 19.42}$	$\mathbf{95.53}\%$	0.24 ± 0.11	$\mathbf{84.67}\%$	2.0%
	PNMPC-N	96.55 ± 49.74	48.27%	0.14 ± 0.003	5.33%	72.67%
Lv. 2	DroQ	-293.13 ± 85.57	-146.57%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-85.42 ± 127.76	-42.71%	$\mathbf{0.002 \pm 0.0002}$	0.0%	100.0%
	SPMPC	152.84 ± 59.40	76.42%	0.15 ± 0.08	49.33%	27.33%
	FPMPC	163.57 ± 44.14	81.78%	0.13 ± 0.07	39.33%	16.67%
	PETS	130.79 ± 59.67	65.39%	0.22 ± 0.09	22.67%	34.0%
	PNMPC	$\mathbf{179.15 \pm 35.28}$	$\mathbf{89.58}\%$	0.23 ± 0.10	$\mathbf{66.67}\%$	$\mathbf{8.0}\%$
	PNMPC-N	92.07 ± 57.29	46.04%	0.14 ± 0.002	5.33%	68.67%
Lv. 3	DroQ	-253.19 ± 42.30	-126.60%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-286.07 ± 42.30	-143.04%	$\mathbf{0.002 \pm 0.0002}$	0.0%	100.0%
	SPMPC	145.29 ± 66.99	72.64%	0.15 ± 0.07	45.33%	30.0%
	FPMPC	161.56 ± 50.97	80.78%	0.14 ± 0.08	43.33%	19.33%
	PETS	129.03 ± 50.07	64.51%	0.22 ± 0.10	18.0%	43.33%
	PNMPC	$\mathbf{180.57 \pm 36.19}$	$\mathbf{90.28}\%$	0.23 ± 0.10	$\mathbf{74.0}\%$	$\mathbf{10.67}\%$
	PNMPC-N	83.08 ± 51.97	41.90%	0.14 ± 0.002	2.0%	74.0%

the efficiency of sparse GP in a larger task area (compared to the position-keeping task) while PNMPC achieved a similar and stable optimization time as in the position-keeping task. In the task with the right trajectory, PNMPC consistently

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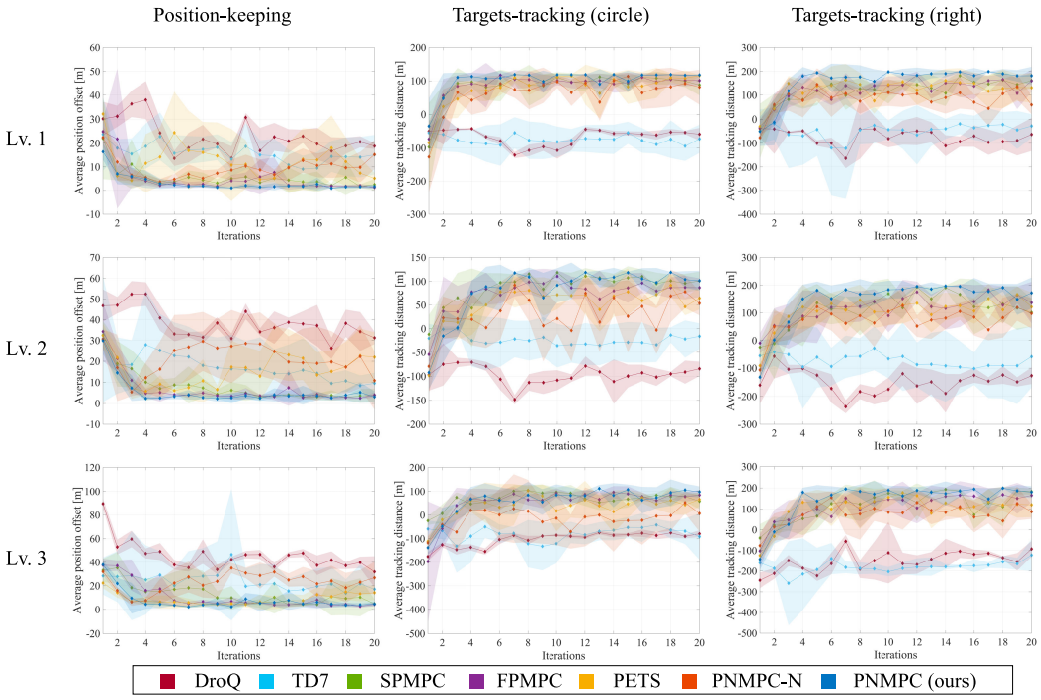


Fig. 4. Learning curves of PNMPC and baselines in three control tasks under different levels of disturbances. Lines and translucent regions represent the mean and standard deviation.

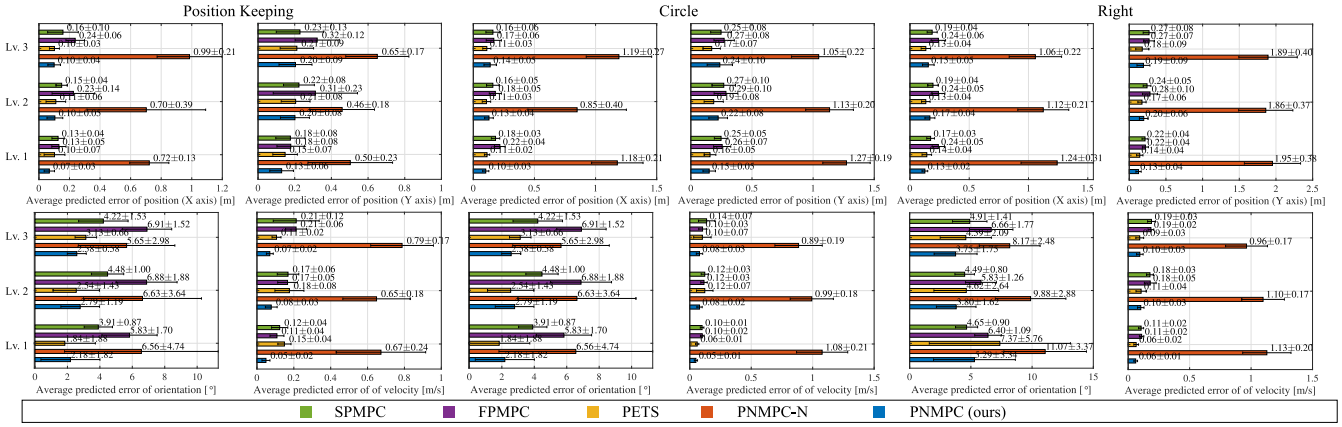


Fig. 5. Average prediction errors of the model learned by PNMPC and other baselines in position-keeping and targets-tracking tasks during the testing.

maintained its superiority and outperformed all other baseline methods. Under the third level of disturbance, the proposed method tracked 10% more distances compared to the sub-optimal approach FPMPC, while improving success rate by 30%. Matching with the deteriorated learning ability shown in Fig. 4, PNMPC-N demonstrated the worst performance compared with all MBRL baselines in the testing. Although the model-free methods DroQ and TD7 had clear advantages in calculation speed, they all failed to guide the USV to complete the given tasks.

PNMPC demonstrated significant advantages over other baselines in terms of success and failure rates, showcasing its robustness in controlling USV in ocean environments with dynamic disturbances. Its comprehensive control capabilities were evident through smaller average offsets and longer driving distances. Furthermore, the consistent excellence over three tasks highlighted its superior generalization ability.

3) Model Accuracy: In this subsection, we evaluated the quality of the probabilistic model learned by PNMPC. The average errors and their standard deviation between the first step prediction by the learned model and the real observation in USV position, orientation and velocity using all model-based baselines were demonstrated in Fig. 5.

In all testing scenarios, PNMPC consistently achieved better prediction accuracy and smaller prediction standard deviation in all dimensions. The model accuracy was maintained even with the increasing disturbances. In comparison, the GP model-based approaches suffered from larger prediction errors caused by the sparse GP model. Although the neural networks-based PETS usually had a similar one-step

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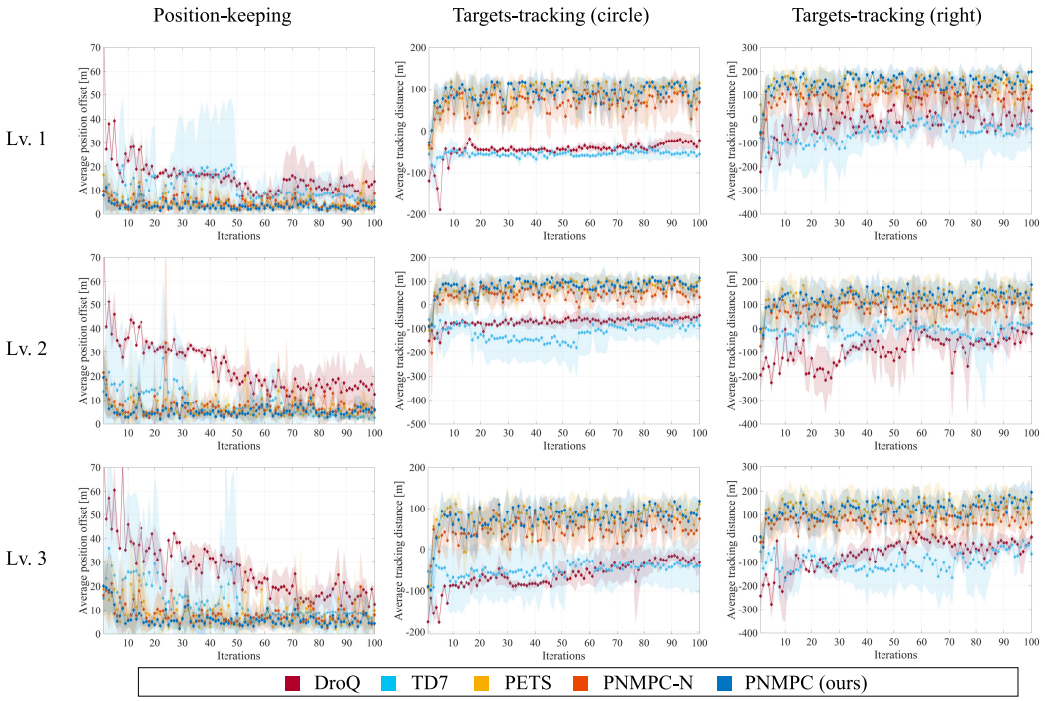


Fig. 6. Learning curves of PNMPC, PETS, TD7 and DroQ in three control tasks with larger sample capacity under different levels of disturbances. Lines and translucent regions represent the mean and standard deviation.

prediction accuracy to PNMPC, it significantly lagged in control performances using multi-step prediction MPC policy. This was due to not only the original loss function that did not consider the USV dynamics in continuous two steps but also the excessive uncertainties in long-term prediction. After disabling the uncertainty propagation, the model prediction error of PNMPC-N significantly increased, which led to the deterioration of the testing performance. Compared to the GP model-based methods SPMPC and FPMPC, the proposed PNMPC significantly enhanced model accuracy. Compared to PETS which modeled the target dynamics by neural networks, our approach substantially improved generalization ability and learning performance by fitting the model in continuous two steps without introducing additional errors in single-step prediction. These results indicated the superiority of PNMPC in the quality of the learned USV model.

C. Evaluation With Large Sample Capacity

In this subsection, the proposed method was evaluated in more challenging ocean environments with more learning iterations and larger sample capability to fully demonstrate its potential as a neural networks-based approach. Based on the three levels of disturbance introduced in Section IV-A, we generated the initial wind direction in a wider range $\psi^w = U(-180, 180)^\circ$. The initial wind velocity was set as $v^w = U(0, 1) \cdot v^w_{max}$ where $v^w_{max} = [2.0, 4.0, 6.0]$ m/s for three levels of disturbance. At each step, the wind direction and velocity were also affected by the random noises $U(-30, 30)^\circ$ and $U(-0.1, 0.1)$ m/s respectively.

The configuration above turned into a substantial expansion of the state space. It significantly increased the task's difficulty, while also being highly valuable for learning more general and robust control strategies in the highly random ocean environment rather than the one with a limited range of disturbances in [28]. To fully explore and learn the tasks under this configuration, we set the number of iterations/rollout during the training to 100, using a total of 5000 to 10000 samples. It is infeasible to employ SPMPC and FPMPC to promptly control USV by the MPC policy under such a large sample set due to the overlarge computational complexity of the GP model discussed in [28]. We compared DroQ, TD7, PETS, PNMPC-N and PNMPC to evaluate the superiority of the proposed method. All algorithmic settings remained consistent with Section IV-A.

1) Learning Performances: The learning curves of the proposed PNMPC and baseline PETS within 100 iterations were compared in Fig. 6. Both PNMPC and PETS demonstrated good learning behaviors in three levels of disturbance. Without uncertainty propagation, PNMPC-N had significant deficiencies in learning stability and finally converged to the worse position offsets and tracking distances. The model-free method DroQ could not learn the tasks within 100 iterations. Although TD7 successfully converged in the position-keeping tasks, it achieved unsatisfactory performances in two tracking tasks. In the position-keeping scenario, PETS frequently experienced drastic degradations in control performance during the training as the environmental disturbances increased. In comparison, our method enjoyed a more robust learning procedure, with a smaller standard deviation in learning curves and finally converged to superior average position offsets. In the target-reaching scenario, PNMPC and PETS had similar convergence behaviors. Their learning curves exhibited larger standard deviations with increasing levels of disturbance.

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TABLE V
TEST RESULT OF PROPOSED METHOD AND PETS WITH MORE LEARNING ITERATIONS IN THE POSITION-KEEPING TASK

Disturbance	Method	Average [m]	Median [m]	Time [s]	Success rate (final)	Success rate (overall)
Lv. 1	DroQ	13.0 ± 8.11	12.43	0.002 ± 0.0004	13.33%	4.44%
	TD7	4.97 ± 2.91	4.57	$\mathbf{0.002 \pm 0.0002}$	84.0%	2.0%
	PETS	5.71 ± 11.67	2.71	0.24 ± 0.08	73.33%	34.67%
	PNMPC	$\mathbf{4.09 \pm 4.94}$	**2.48**	0.24 ± 0.08	**85.33%**	**42.0%**
	PNMPC-N	5.38 ± 8.33	2.59	0.21 ± 0.01	74.67%	37.33%
Lv. 2	DroQ	21.37 ± 17.23	25.61	0.002 ± 0.0004	20.0%	0.0%
	TD7	$\mathbf{3.79 \pm 2.48}$	3.31	$\mathbf{0.002 \pm 0.0003}$	**93.33%**	20.0%
	PETS	7.54 ± 11.09	3.75	0.24 ± 0.08	72.0%	18.67%
	PNMPC	6.31 ± 8.13	**3.10**	0.25 ± 0.09	76.0%	**26.67%**
	PNMPC-N	9.39 ± 13.43	4.99	$\mathbf{0.21 \pm 0.01}$	59.33%	12.67%
Lv. 3	DroQ	30.96 ± 22.90	32.73	0.002 ± 0.0003	0.0%	0.0%
	TD7	73.96 ± 46.26	75.48	$\mathbf{0.002 \pm 0.0002}$	0.0%	0.0%
	PETS	8.69 ± 20.03	3.91	0.24 ± 0.09	70.0%	26.0%
	PNMPC	$\mathbf{5.73 \pm 7.73}$	**3.53**	0.25 ± 0.09	**74.67%**	**27.33%**
	PNMPC-N	10.32 ± 14.19	5.62	0.21 ± 0.01	54.0%	6.0%

TABLE VI
TEST RESULT OF PROPOSED METHOD AND PETS WITH MORE LEARNING ITERATIONS IN THE TARGETS-TRACKING TASK (CIRCLE)

Disturbance	Method	Average complete distance [m]	Average complete ratio	Time [s]	Success rate	Failure rate
Lv. 1	DroQ	-38.41 ± 33.32	-32.0%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-103.43 ± 12.51	-86.19%	$\mathbf{0.002 \pm 0.0002}$	0.0%	100.0%
	PETS	98.69 ± 26.59	82.24%	0.20 ± 0.07	62.0%	19.33%
	PNMPC	$\mathbf{104.16 \pm 23.44}$	**85.13%**	0.22 ± 0.09	**73.33%**	**10.0%**
	PNMPC-N	68.45 ± 77.31	57.04%	0.15 ± 0.004	34.0%	40.0%
Lv. 2	DroQ	-94.23 ± 47.66	-78.52%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	TD7	-146.56 ± 56.10	-122.13%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	PETS	80.91 ± 41.43	67.43%	0.21 ± 0.09	45.33%	39.33%
	PNMPC	$\mathbf{95.36 \pm 32.64}$	**79.47%**	0.23 ± 0.11	**60.67%**	**22.0%**
	PNMPC-N	58.96 ± 44.13	49.14%	0.15 ± 0.004	19.33%	63.33%
Lv. 3	DroQ	-37.35 ± 29.21	-31.12%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	TD7	-40.14 ± 68.37	-33.45%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	PETS	85.19 ± 36.89	71.0%	0.22 ± 0.10	42.67%	34.0%
	PNMPC	$\mathbf{93.98 \pm 30.43}$	**78.32%**	0.22 ± 0.10	**51.33%**	**24.67%**
	PNMPC-N	57.63 ± 45.72	48.02%	0.15 ± 0.004	19.33%	64.0%

Unlike the significant differences in learning curves between PETS and PNMPC shown in Fig. 4, Section IV-B, they had similar learning curves in the scenarios where some observed states came from highly random distributions. We believed this phenomenon was caused by the neural networks' characteristics: they are highly effective in learning data with large distributions without being negatively affected by the excessive update within a relatively local sample space.

2) Testing Performances: The control performances of TD7, DroQ, PETS, PNMPC and PNMPC-N in the testing were compared in Tables V, VI and VII to investigate the generalization capability of our method after the 100 interactions' training. The bold font indicated the best performance in the corresponding term. Despite the similar learning curves of FPMPC and PETS in Fig. 6, the proposed method consistently showed advantages in the testing while PETS failed to demonstrate sufficient generalization capability and stability. In the position-keeping task, PNMPC outperformed PETS in all metrics related to control performances. Under the first level of disturbance, it enjoyed over 10% improvement in success rate (final) while reducing the average position offset by 28%. Although the two methods suffered decreases in success rate under the strongest level of disturbance, the proposed method still maintained over 4% advantage and achieved a 34% lower position offset than PETS. Without the support of uncertainty propagation, PNMPC-N achieved close control performance and success rate to PETS after a long training process. As the disturbance level increased, the performance gap between PNMPC-N and PETS also widened. For the model-free approaches, DroQ could not learn this task

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TABLE VII
TEST RESULT OF PROPOSED METHOD AND PETS WITH MORE LEARNING ITERATIONS IN THE TARGETS-TRACKING TASK (RIGHT)

Disturbance	Method	Average complete distance [m]	Average complete ratio	Time [s]	Success rate	Failure rate
Lv. 1	DroQ	-38.35 ± 103.05	-19.18%	0.002 ± 0.0004	0.0%	93.33%
	TD7	-73.97 ± 161.26	-36.99%	$\mathbf{0.002 \pm 0.0002}$	2.0%	90.67%
	PETS	135.89 ± 57.46	67.87%	0.22 ± 0.10	34.0%	40.67%
	PNMPC	$\mathbf{159.33 \pm 53.27}$	$\mathbf{79.67\%}$	0.23 ± 0.10	$\mathbf{54.67\%}$	$\mathbf{23.33\%}$
	PNMPC-N	110.33 ± 61.61	55.16%	0.15 ± 0.006	16.67%	52.0%
Lv. 2	DroQ	-84.18 ± 96.05	-42.09%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	TD7	17.91 ± 37.45	-9.0%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	PETS	120.46 ± 66.25	60.23%	0.22 ± 0.09	27.33%	48.67%
	PNMPC	$\mathbf{155.53 \pm 50.28}$	$\mathbf{77.77\%}$	0.23 ± 0.10	$\mathbf{45.33\%}$	$\mathbf{23.33\%}$
	PNMPC-N	91.07 ± 65.83	45.53%	0.15 ± 0.007	12.0%	67.33%
Lv. 3	DroQ	-55.66 ± 26.17	-27.83%	0.002 ± 0.0004	0.0%	100.0%
	TD7	-39.23 ± 43.25	-19.61%	$\mathbf{0.002 \pm 0.0003}$	0.0%	100.0%
	PETS	120.46 ± 66.25	60.23%	0.22 ± 0.09	27.33%	48.67%
	PNMPC	$\mathbf{152.43 \pm 50.52}$	$\mathbf{76.21\%}$	0.23 ± 0.10	$\mathbf{41.33\%}$	$\mathbf{26.0\%}$
	PNMPC-N	86.89 ± 62.03	43.45%	0.15 ± 0.007	4.67%	70.67%

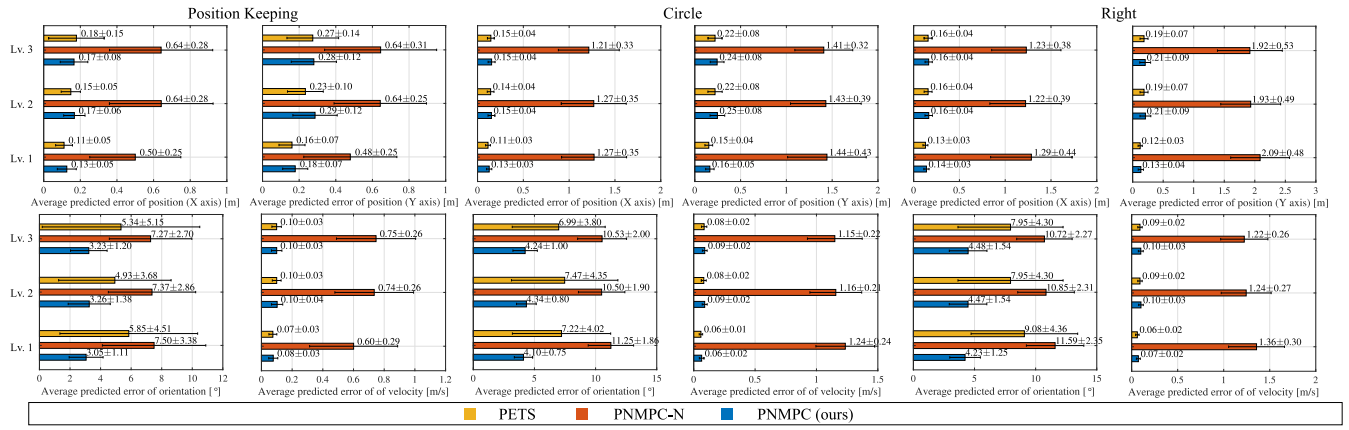


Fig. 7. Average prediction errors of the model learned by PNMPC and PETS in position-keeping and targets-tracking tasks with large sample capacity during the testing.

and achieved very low success rates in all levels' disturbances. While TD7 performed very well under the disturbances of the first two levels, its control ability declined rapidly against stronger disturbances. It was entirely incapable of completing the task under the disturbances of level three.

In the targets-tracking scenario with the circle trajectory, the proposed PNMPC also achieved control performance that surpassed the baseline PETS. On average, it completed 8% more tracking distance under three levels of disturbance while significantly improving the quality of task completion: the success rate increased by about 9% and the failure rate decreased by about 10%. In the targets-tracking scenario with the right trajectory that addressed long-term and high-speed driving, PNMPC demonstrated significant superiority over PETS. As the level of disturbance increased, our method improved the tracking distance by 12%, 17% and 16% and enhanced the success rate of the task by 20%, 18% and 14%. Disabling the uncertainty in the probabilistic neural networks, PNMPC-N had the worst performances in both circle and right trajectories. Without the guidance of the model, model-free RL failed to learn this task, DroQ and TD7 could not even track the first target under all levels of disturbances. The results above demonstrated the superior control performance and generalization capability of the proposed method in learning USV control tasks with a large sample capacity.

Regarding computational complexity, the average decision-making time for all methods remained consistent with their performances in Section IV-B, irrespective of increased sample capacity. Although PNMPC's uncertainty propagation is more complex than PETS due to additional dropout units, we consider the slightly longer optimization time of approximately 0.01 s acceptable in practice. In contrast, all GP-based approaches were infeasible for these tasks, as the optimization time per step with over $10,000$ samples could exceed 10 seconds. While the decision-making time of model-free approaches DroQ and TD7 was about 1% of our method, they demonstrated significant shortcomings in both control performance (average offset and driving distance) and robustness (success and failure rates). These disadvantages became more pronounced as environmental disturbances intensified,

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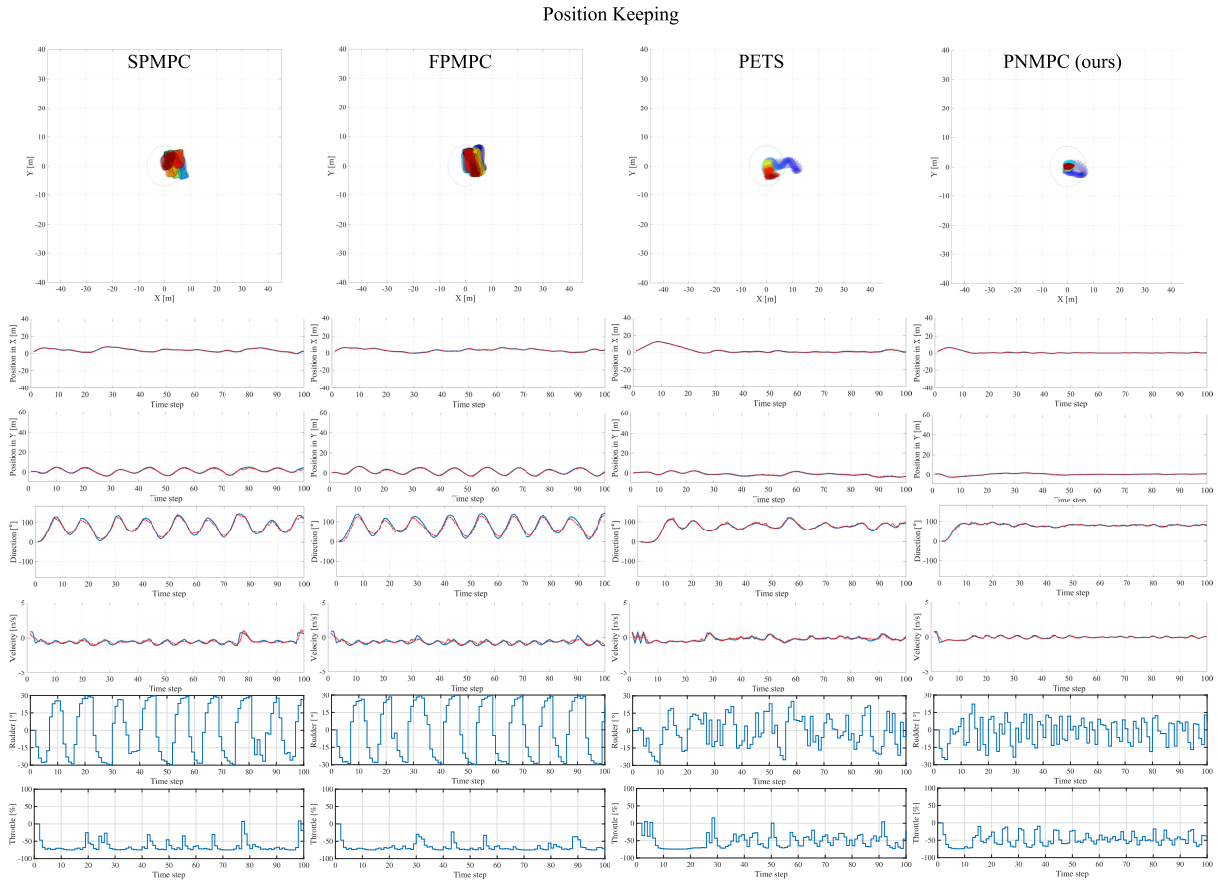


Fig. 8. One test rollout with state and action trajectories of all compared approaches in position-keeping under the third level of disturbance.

rendering these methods nearly incapable of completing tasks under the highest level of disturbances. Consequently, we believed that the trade-off between computation time and performance in our method was reasonable.

3) Model Accuracy: Figure 7 compared the average error and standard deviation between the first step prediction in the MPC policy and the observed state using the models learned by PNMPC, PNMPC-N and PETS during the testing. It can be observed that as the sample capacity increased, the large prediction errors in USV position and velocity were partially alleviated even under more random and challenging disturbances compared with the results in Section IV-B. Both PETS and PNMPC demonstrated reasonable prediction accuracy in these states. On the other hand, PNMPC demonstrated significant advantages over PETS in both accuracy and stability when predicting the orientation of USV which was strongly affected by the ocean disturbances. It consistently enjoyed far smaller prediction errors and lower standard deviation under various levels of disturbances. We believe these advantages mainly come from the proper uncertainty representation and propagation in the proposed method. Once the uncertainty was disabled, the model prediction error of PNMPC-N quickly increased and led to a control performance inferior to PETS.

D. Case Study

In this subsection, a case study of position-keeping and targets-tracking scenarios using all compared approaches was conducted to illustrate the control behavior learned by the proposed method. After learning in 20 iterations with the same random seed, all approaches were tested in one rollout with the same random distances. The testing rollout including the USV state and action trajectories were shown in Figs. 8, 9 and 10. The boat shapes indicated the states of USV at time steps 1 to 100 (blue to red). The yellow stars indicated tracking targets. The predicted values and the observed states were shown in red and blue respectively.

In the position-keeping task shown in Fig. 8, all compared approaches were able to maintain the USV position within a range of 7 meters to the initial position at the last step. However, compared to the GP model-based approaches SPMPC and FPMPC which resulted in a wider range of USV position and orientation, the proposed PNMPC consistently kept the position of USV as close to the initial position as possible, except for necessary adjustments of orientation. The proposed probabilistic neural networks accurately predicted the orientation of the USV, which contributed to a smooth trajectory of orientation with the smallest fluctuation range, helping the USV to maintain its position under disturbances. Meanwhile, the neural networks-based PETS could not achieve similar control performances. Compared to other GP model-based baselines, the neural networks model in PETS sacrificed model accuracy. It deviated far from the initial position with larger position offsets than our method. In terms of the states and actions trajectories, thanks to its superior prediction accuracy

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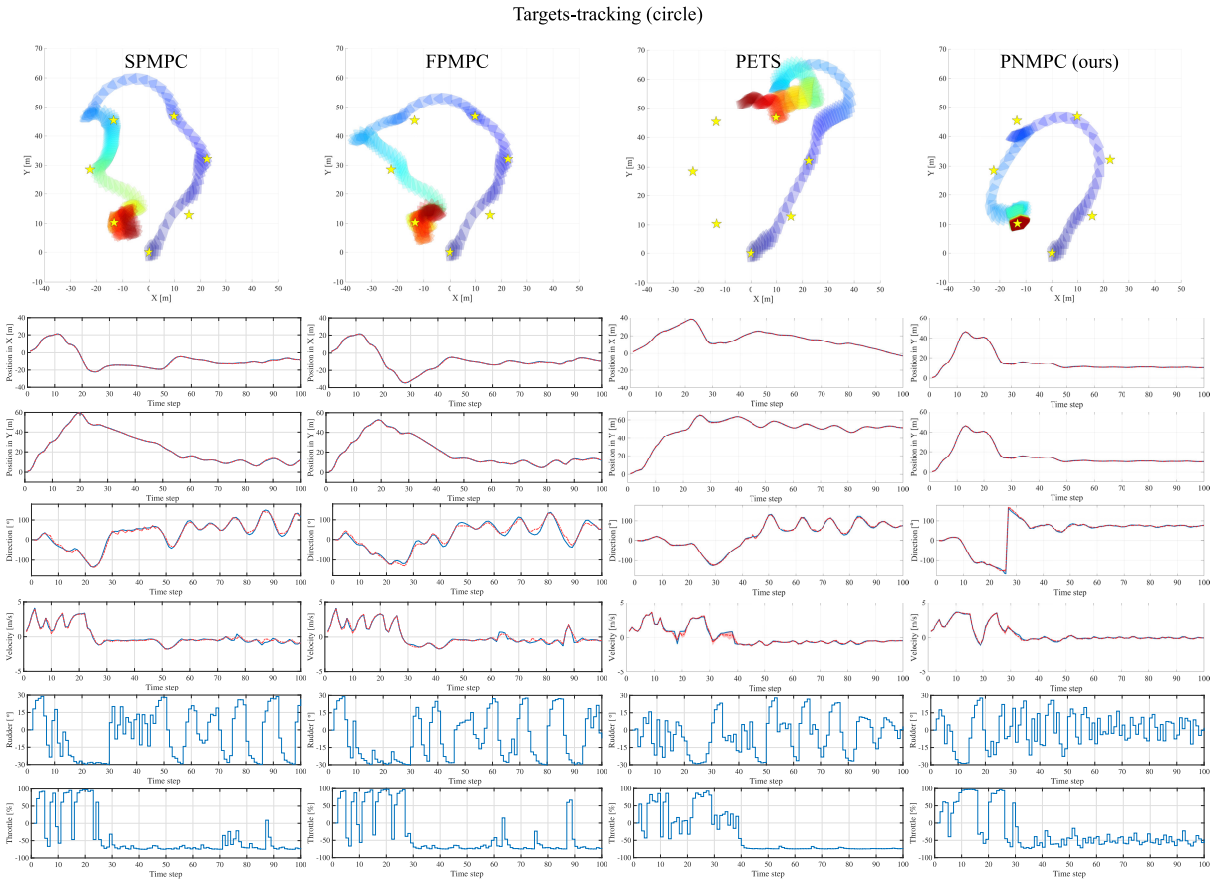


Fig. 9. One test rollout with state and action trajectories of all compared approaches in targets-tracking (circle) under the third level of disturbance.

of USV orientation than the GP model-based approaches, PNMPC effectively controlled the USV heading while avoiding the fluctuation of orientations. Additionally, with a longer prediction horizon in MPC policy, PNMPC always conducted smaller rudder and throttle commands, resulting in more precise control behaviors.

In the targets-tracking scenario with the circle trajectory as shown in Fig. 9, PNMPC demonstrated superior and effective control behavior compared with other baselines. With redundant USV trajectories, SPMPC and FPMPC remained in unstable position-keeping after tracking all targets. PETS was unable to complete the task at all. The large predicted error of velocity in the early stage caused the USV to move too fast when tracking the second target. The subsequent violent swaying in orientation further prevented the USV from resetting, ultimately leading to the failure of the task. As a comparison, PNMPC quickly tracked all targets and stabilized at the final target within approximately 40 steps. The effective uncertainty propagation in our method resulted in prompt feedback to the changing disturbances using the MPC policy and therefore contributed to smooth and seamless control trajectories of both USV position and orientation. The proper cooperation of the rudder and engine throttle drove the USV to quickly pass through all targets at a relatively high speed and finally stop steadily at the last target point.

According to Fig. 10, PNMPC had similar advantages in the long-distance tracking task (right trajectory). Based on the traditional probabilistic neural networks model, the MPC policy of PETS failed to promptly respond to the changing disturbances, resulting in the USV stalling near the fourth target. In comparison to the GP model-based approaches that excessively adjusted the USV velocity to sequentially track multiple targets due to the large predicted error of direction and velocity, our method significantly reduced the frequency of challenging throttle and contributed to superior efficiency and stability in the tracking task. Once the final target was reached, PNMPC properly controlled the rudder and throttle, stably holding the USV's position.

E. Discussion of PNMPC and Its Implementation

The experimental results above demonstrated the significant advantages of PNMPC over current state-of-the-art model-free approaches in USV control scenarios. The agent of model-free RL based on Actor-Critic networks attempts to sacrifice the stability of the current system in pursuit of maximizing the accumulated rewards in the long term, leading to loss of control in disturbed environments. In contrast, PNMPC achieved a superior balance between accumulated rewards and system stability in a short horizon via the MPC controller, making it more suitable for unmanned ship control. However, it faces a heavier computational burden compared to DroQ and TD7, while its performance is highly dependent on the MPC horizon and the reliability of the learned model.

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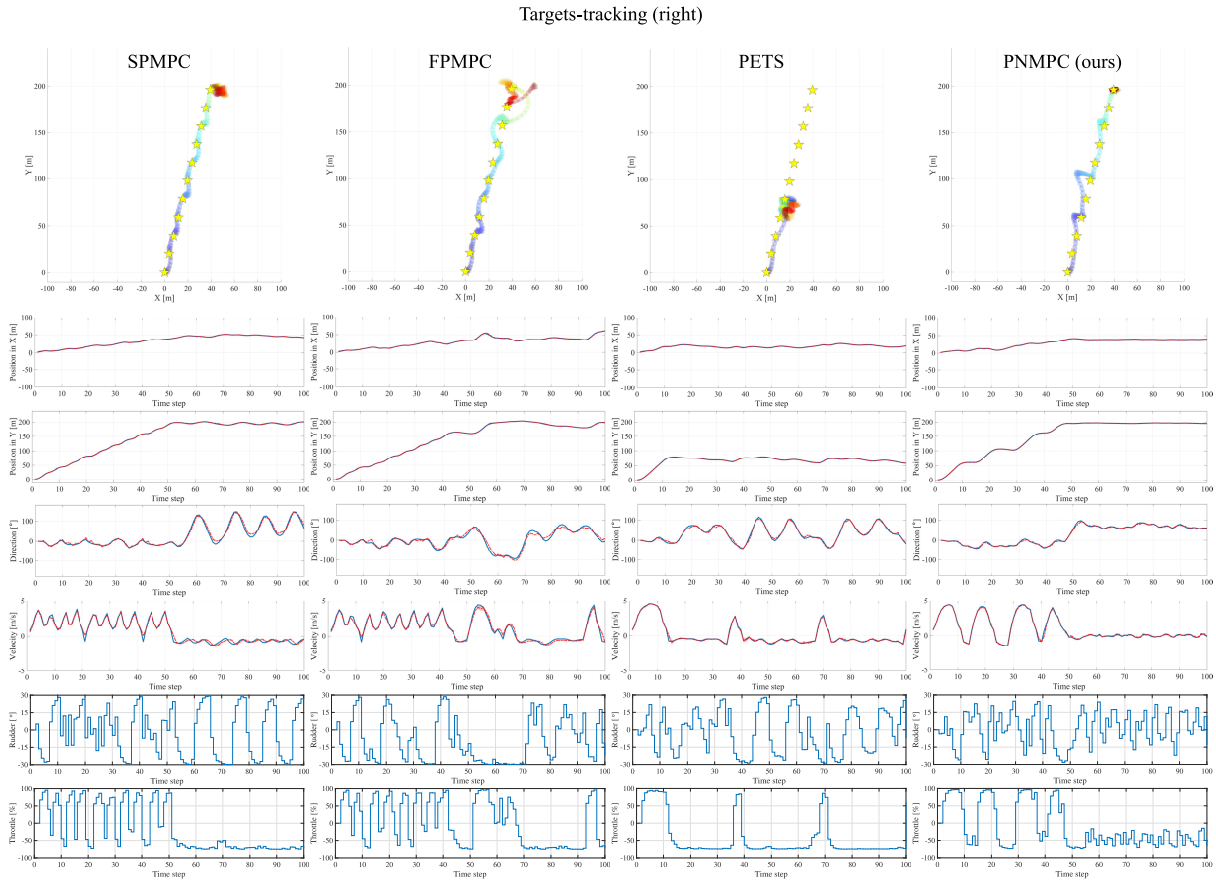


Fig. 10. One test rollout with state and action trajectories of all compared approaches in targets-tracking (right) under the third level of disturbance.

For applications, PNMPC can be directly implemented in a real-world USV system following [28] while achieving a superior control frequency independent of the size of samples. Although the stability of probabilistic neural networks has been less discussed in related works [35], [36], the proposed method provides a practical mechanism to alleviate unknown machine faults and unexpected system behaviors in USV control scenarios: these anomalies may lead to an increase in the standard deviation of the model predictions, signaling the system to promptly halt autonomous decision-making, thereby preventing potentially dangerous situations. In non-ideal scenarios, such as when observed states are missing or corrupted by noise, related MBRL approaches GP-MPC, FPMPC, and PETS often predict states with inaccurate distributions, leading to accumulated bias in subsequent steps. As a comparison, our method leverages a two-step loss function in model updating and a mean-based propagation in decision-making to tackle this issue, enhancing the stability of control behavior.

In terms of scalability, employing other Bayesian neural networks in the proposed method was straightforward. However, traditional Bayesian networks mainly designed for supervised learning usually had numerous parameters and struggled to efficiently express uncertainties [43], [44]. We believed the probabilistic neural networks in PNMPC that focused on stably and efficiently modeling system dynamics were more suitable for the real-time control of USV. Algorithmically, integrating the probabilistic model of PNMPC with the Actor-Critic networks to balance the long-term reward and the current system stability is also a meaningful direction.

V. Conclusion

This article proposed PNMPC, a novel probabilistic MBRL approach specialized for USV control. It tackled the computational efficiency issue present in current GP model-based MBRL USV approaches. Within one MBRL framework, PNMPC utilized deep neural networks to properly model the uncertain dynamics of the USV from a probabilistic perspective and incorporated an MPC policy to robustly control the USV against various external disturbances. A novel loss function was designed to emphasize the characteristics of USV dynamics in continuous time steps to properly train the probabilistic neural networks model. The effectiveness of the proposed in modeling USV dynamics and learning control strategies was evaluated across various USV scenarios. Compared to recent GP model-based MBRL approaches, PNMPC offers computational complexity that is independent of sample capacity and plans more optimal USV driving trajectories. Compared with existing work PETS based on neural networks model, PNMPC achieves higher model prediction accuracy and more responsive control behavior against external disturbances, breaking through limitations of control capability traditionally encountered with probabilistic neural networks in engineering scenarios. This work expands the potential

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of probabilistic neural network model-based reinforcement learning towards a fully autonomous USV.

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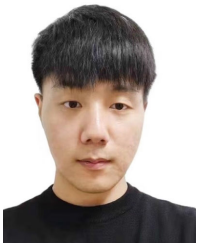
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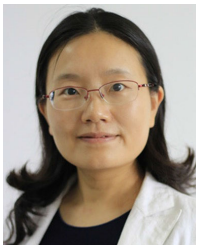
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